

Homework 8: Normalization

Assigned on: Mar 14

Due on: Mar 20

Question 1

For the following table, draw a dependency diagram. Indicate any partial or transitive dependencies on your diagram. Then, decompose this table into tables in third normal form. The primary key is underlined.

<u>Invoice No.</u>	<u>Product No.</u>	Sale Date	Prod Desc	Vend Code	Vend Name	Qty Sold	Prod Price
<u>211347</u>	<u>AAE33</u>	3/25/98	Drill Bit	211	Black&Decker	3	2.95
<u>211347</u>	<u>AR456</u>	3/25/98	Drill	211	Black&Decker	1	29.99
<u>211348</u>	<u>AR444</u>	4/5/99	Screw	133	Acme	55	.17

Question 2

For the following table, draw a dependency diagram. Indicate any partial or transitive dependencies on your diagram. Then, decompose this table into tables in third normal form.

<u>StuID</u>	StuName	StuMajor	DeptID	DeptName	DeptPh	College	AdvName	AdvOffice	AdvBldg	AdvPh	StuGPA	StuHrs	StuClass
<u>111</u>	Smith	Biology	BIOL	Biology	2363333	A&S	Baker	LS113	Biol	2362222	3.01	65	Junior
<u>123</u>	Jones	Biology	BIOL	Biology	2363333	A&S	Williams	LS114	Biol	2362333	3.33	90	Junior
<u>122</u>	Howard	Acct	ACCT	Accounting	2364444	Bus	Boes	BA333	BA	2363232	3.11	20	Freshman

Question 3

For the following table, draw a dependency diagram then, decompose this table into tables in first, second and third normal form.

<u>PET ID</u>	PET NAME	PET TYPE	PET AGE	OWNER	VISIT DATE	PROCEDURE
246	ROVER	DOG	12	SAM COOK	JAN 13/2002 MAR 27/2002 APR 02/2002	01 - RABIES VACCINATION 10 - EXAMINE and TREAT WOUND 05 - HEART WORM TEST
298	SPOT	DOG	2	TERRY KIM	JAN 21/2002 MAR 10/2002	08 - TETANUS VACCINATION 05 - HEART WORM TEST
341	MORRIS	CAT	4	SAM COOK	JAN 23/2001 JAN 13/2002	01 - RABIES VACCINATION 01 - RABIES VACCINATION
519	TWEEDY	BIRD	2	TERRY KIM	APR 30/2002 APR 31/2002	20 - ANNUAL CHECK UP 12 - EYE WASH

Question 4

Given the following relation and example data, List the functional dependencies and normalize this relation into BCNF.

<u>PartNumber</u>	Description	Supplier	SupplierAddress	Price
10010	20 GB Disk	Seagate	Cupertino, CA	\$100
10010	20 GB Disk	IBM	Armonk, NY	\$90
10220	256 MB RAM card	Kensington	San Mateo, CA	\$220
10220	256 MB RAM card	IBM	Armonk, NY	\$290
10220	256 MB RAM card	Sun Microsystems	Palo Alto, CA	\$310
10440	17" LCD Monitor	IBM	Armonk, NY	\$2,100